MATH 2123 (Mathematics-III)

Past Questions 2015-2025

Questions Only - Combined Original Exam Scans

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management

Compiled from the question-paper images supplied in this chat.
No generated solution-guide pages are included.

Updated version: 2022 page 3 has now been included.

Contents

Only orientation correction and mild readability enhancement were applied to the original scans.

2025

MATH 2123 - Question Paper

2 supplied exam pages

Khulna University of Engineering & Technology

Department of Building Engineering and Construction Management

B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2025

Math 2123

(Mathematics-III)

Full Marks: 210

Time: 3 hrs

N.B.
- i) Answer any three questions from each section in separate script.
- ii) Figures in the right margin indicate full marks.
- iii) Necessary table/ graphs/ charts (if any)

Section – A

1. (a) Define the following matrices with examples: (12)
(i) Lower triangular matrix, (ii) Symmetric matrix, (iii) Nilpotent matrix, (iv) Idempotent matrix.

(b) Test whether the following matrix A is idempotent or not. (04)

$$A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$$

(c) Express the matrix A as the sum of a symmetric and skew-symmetric matrices. Where, (08)

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 5 & 0 & 3 \\ 2 & 4 & 7 \end{bmatrix}$$

(d) Solve the following system of linear equations by the matrix method: (11)

$$4x - 5y + z = 2$$
$$3x + y - 2z = 9$$
$$x + 4y + z = 5$$

2. (a) Does inverse exist for all matrices? Test whether the following matrix is invertible or not. If possible, find the inverse of the following matrix using elementary transformation: $\begin{bmatrix} 1 & 20 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$ (13)

(b) Determine the values of a and b for which the system 3x-2y+z=b, 5x-8y+9z=3, 2x+y+az= -1 (i) has a unique solution, (ii) has no solution, and (iii) has infinite solutions. (12)

(c) Define linearly dependent and independent vectors. Examine the following vectors for linear dependence: $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$. (10)

3. (a) Define rank of a matrix. Reduce the following matrix A to Echelon form, then to canonical form and then to normal form and hence find its rank: where, (20)

$$A = \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

(b) Define eigen values and eigen vector. Find the eigen values and eigen vector corresponding one of the eigen values for the matrix (15)

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

4. (a) A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$, $z = 2\sin 3t$, where t is the time. (12)

(i) Determine the magnitude of the velocity and acceleration at t = 0.

(b) Find the work done in moving an object along a vector $\vec{r} = 3\hat{\imath} + \hat{\jmath} - 5\hat{k}$ if the applied force is $\vec{F} = 2\hat{\imath} - \hat{\jmath} - \hat{k}$. (10)

(c) State Cayley-Hamilton theorem. Verify it for the matrix $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and hence find A^{-1}. (13)

Section – B

5\. (a) Write down the sufficient conditions for existence of Laplace transforms. Find the Laplace transform of (13)

(i) $F(t) = (t+2)\, e^{-3t} \sin 5t$.

(ii) $G(t) = \begin{cases} t & \text{for } 0 < t < 2 \\ 0 & \text{for } 2 < t < 4 \end{cases}$ $\quad and \ G(t+4) = G(t)$.

(unit step function)

(b) Define unit step function. The load distribution on a beam is: (10)

$$p(t) = \begin{cases} 5t, & 0 < t < 4 \\ 20, & t \geq 4 \end{cases}$$

Write P (t) in terms of unit step function and determine the Laplace transform.

(c) Determine the inverse Laplace transform of (i) $f(s) = \frac{e^{-5s}}{(s-2)^4}$ (12)

(ii) $g(s) = \frac{2s^2-4}{(s+1)(s-2)(s-3)}$

6\. (a) State the convolution theorem. Use the convolution theorem to evaluate $\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2(s-2)}\right\}$. (11)

(b) The displacement of a building during earthquake satisfies: $x'' + 2x' + x = e^{-t}$ with x (0) = 0, x' (0) = 1. Find the displacement response using Laplace transform. (13)

(c) The temperature distribution in a reinforced concrete structure is: (11)

$$T(x,y,z) = x^2 y\, e^{-z} + 3xy^2 + z^3.$$

Find (i) Magnitude of temperature gradient at (1, 2, 0).

(ii) Direction of maximum heat increase.

7\. (a) The wind velocity around a high-rise building is: $\vec{V} = (x^2y + yz)\hat{\imath} + (xy^2 + xz)\hat{\jmath} + (yz^2 + x^2z)\hat{k}$. Determine whether the airflow has expansion or compression at point (1, 2, 1). (11)

(b) Air velocity field in a smart building system is: $\vec{V} = (x^3 + yz)\hat{\imath} + (y^3 + xz)\hat{\jmath} + (z^3 + xy)\hat{k}$. Check whether the velocity field is irrotational or not. If exists, find the corresponding velocity potential. (12)

(c) A force field in a construction system is $\vec{F} = (2xy + z^2)\hat{\imath} + (x^2 + 2yz)\hat{\jmath} + (2xz + y^2)\hat{k}$. Check whether this field is conservative or not. Find the work done in moving an object in this field from (1, 0, 0) to (2, 1, 1). (12)

8\. (a) Evaluate $\iint_s \vec{A} \cdot \vec{n}\, ds$, where $\vec{A} = 18z\,\hat{\imath} - 12\,\hat{\jmath} + 3y\,\hat{k}$ and s is that part of the plane $2x + 3y + 6z = 12$, which is located in the first octant. (12)

(b) Evaluate $\iiint_v \vec{F}\, dv$ where $\vec{F} = 2xz\,\hat{\imath} - x\hat{\jmath} + y^2\hat{k}$ and v is the region bounded by the surfaces x = 0, y = 0, y = 6, z = x^2, z = 4. (11)

(c) Suppose a force field is given by $\vec{F} = (2x - y + z)\hat{\imath} + (x + y - z^2)\hat{\jmath} + (3x - 2y + 4z)\hat{k}$. Find the work done in moving a particle once around a circle C in the xy-plane with its centre at the origin and a radius of 3. 12

2024

MATH 2123 - Question Paper

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(60)

Khulna University of Engineering & Technology

Department of Building Engineering and Construction Management

B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2024

Math 2123

(Mathematics-III)

Full Marks: 210

Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable value for any missing data.

Section – A

1. (a) A freshly cast concrete slab has a temperature field: $T(x,y,z) = 40 - 0.1x^2 - 0.05y^2 - 0.2z$ (°C) where x, y, z are in meters. Find (11)
   (i) the temperature gradient at (2, 3, 0.5)
   (ii) the direction of maximum heat flow.

   (b) Find the constants a, b, c so that $\vec{F} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{k}$ is irrotational and hence find the corresponding scalar potential. (12)

   (c) Find the value of the constant a and b such that the surface $ax^2 - byz = (a+2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point (1, -1, 2). (12)

2. (a) Find the work done in moving a particle in the force field $\vec{F} = 3x^2\,\hat{\imath} + (2xz - y)\hat{\jmath} + z\hat{k}$ along (i) the straight line from (0 ,0 ,0) to (2, 1, 3), (12)
   (ii) the curve defined by $x^2 = 4y$, $3x^3 = 8z$ from $x = 0$ to $x = 2$

   (b) Evaluate $\int_s \vec{F}.\hat{\eta}\, ds$ where s is the surface $x + y + 2z = 4$ cut by the planes $x = 0$, $y = 0$, $z = 0$ and $\vec{F} = (2x^2 - 3z)$ and $\hat{\imath} - 2xy\,\hat{\jmath} - 4x\,\hat{k}$. (12)

   (c) Let, $\vec{F} = 2xz\,\hat{\imath} - x\hat{\jmath} + y^2\hat{k}$. Evaluate $\iiint_v \vec{F}\,dv$, where v is the region bounded by the surfaces $x = 0$, $y = 0$; $y = 6$, $z = x^2$, $z = 9$. (11)

3. (a) Find $L\{F(t)\}$, where $F(t) = \begin{cases} 3t \;;\; 0 < t < 2 \\ 6 \;;\; 2 < t < 4 \end{cases}$ and $F(t + 4) = F(t)$ (11)

   (b) Evaluate $\int_0^\infty t^3 e^{-t} \sin t\; dt$. (12)

   (c) Find the Laplace transform of $\frac{\sin at}{t}$. Does the Laplace transform of $\frac{\cos at}{t}$ exist? Justify. (12)

4. (a) State the convolution theorem. Use the convolution theorem to find: (12)
   $L^{-1}\left\{\frac{1}{(S+1)\,(S^2+1)}\right\}$

   (b) Find the inverse transform of $\frac{S}{S^2 - 2S + 35}$ (10)

   (c) Using Laplace transform, solve $Y'' + 9Y = \cos 2t$ if $Y(0) = 1$, $Y\left(\frac{\pi}{2}\right) = -1$ (13)

Section – B

5. (a) Define with an example and two properties : (i) Orthogonal matrix, (ii) Elementary and (iii) Nonsingular matrix . (9)

(b) Are vectors [0 2 3 4], [2 3 0 1] , and [4 8 3 6] linearly independent? If so, why? If possible, express an independent vector as a linear combination of dependent vectors. (13)

(c) What is meant by invertible matrix? If exists, find the inverse of the following matrix A by elementary row transformations where, $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ (13)

6. (a) What do you mean by the rank of a matrix? Reduce the matrix $\begin{bmatrix} 0 & 3 & -1 \\ 1 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$ to its Row echelon form, Canonical form, and Normal form. If possible, find its inverse matrix as a by product of the process to obtain its normal form. Also determine its rank. (18)

(b) What is the value of K for which the following system has : (i) an unique solution, (ii) many solutions, (iii) no solution? If possible, find the unique solution of the following system using the Gauss elimination method (elementary row operations): (17)

$$\begin{bmatrix} K & 3 & 2 \\ 0 & 6 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} K \\ 1 \\ 1 \end{bmatrix}.$$

7. (a) Given $-2x + y + 3z = 0$, $2y + z = 0$, $x - 4y + Tz = 0$, What is the value of T so the system of equations has: (i) trivial solution (ii) nontrivial solution? If possible, find the nontrivial solution of the system for some value of T, and express the nontrivial solution in parametric form. (15)

(b) Write down 2 properties for any eigensystem. Find the eigenpair and corresponding eigenspace of A, Where $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Moreover, mention algebraic multiplicity and geometric multiplicity of the above problem. Also, find the eigenvalues and corresponding eigenvectors of the matrix $A - 4I$, where I is the identity matrix. (20)

8. (a) A space curve is defined by the position vector $5\cos t\,\hat{\imath} + 5\sin t\,\hat{\jmath} - \hat{k}$. Find the unit tangent vector $\hat{T}$, normal vector $\hat{N}$, and binormal vector $\hat{B}$. Calculate the radius of curvature, acceleration, and the tangent & normal components of acceleration. Additionally, determine the trihedral plane that contains the binormal vector $\hat{B}$ and normal vector $\hat{N}$ at $t = \frac{\pi}{2}$ (28)

(b) A force $\overline{F} = -t\,\hat{\imath} + t^2\,\hat{\jmath}$ is acting on a moving particle with a mass of 2 units. Find its velocity such that the velocity at $t = 0$ is $5\hat{\imath}$. (07)

$\hat{T} = \dfrac{\frac{dA}{dt}}{\text{its value}}$, $\hat{N} = \dfrac{\frac{d\hat{T}}{dt}}{\text{Value}}$,

$\hat{B} = \hat{T} \times \hat{N}$

$K = \dfrac{|\hat{T}'|}{|R'|}$

$A_t = a \cdot \hat{T}$ = tangential component of acceleration

$\dfrac{1}{K} = \rho$

$\sqrt{a^2 - A_t^2} = \boxed{A_n}$

2023

MATH 2123 - Question Paper

2 supplied exam pages

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2023
Math 2123
(Mathematics - III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Find the constants a, b, c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at $(1, 2, -1)$ has a maximum of magnitude 64 in a direction parallel to the z axis. (11)

(b) A fluid motion is given by $\vec{v} = (y\sin z - \sin x)\hat{\imath} + (x\sin z + 2yz)\hat{\jmath} + (xy\cos z + y^2)\hat{k}$. Is the motion irrotational? If so, find the corresponding velocity potential. (12)

(c) Find the values of constants λ and μ so that the surfaces $\lambda x^2 - \mu yz = (\lambda + 2)x, 4x^2y + z^3 = 4$ intersect orthogonally at the point $(1, -1, 2)$. (12)

2. (a) If $\vec{A} = (y - 2x)\hat{\imath} + (3x + 2y)\hat{\jmath}$, compute the circulation of A about a circle C in the xy plane with center at the origin and radius 2, if C is traversesd in the positive direction. (11)

(b) Evaluate $\iint_S \vec{A}.\vec{n}\, ds$, where $\vec{A} = z\hat{\imath} + x\hat{\jmath} - 3y^2z\hat{k}$ and s is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$. (12)

(c) Find the Laplace transform of the following functions: (12)
(i) $F(t) = \frac{e^{-t}\sin t}{t}$
(ii) $G(t) = te^{-3t}\sin 2t$

3. (a) Evaluate $\iiint_V \overline{\nabla.F}\, dv$, where V is the closed region bounded by the planes $x = 3, y = 2, z = 1$ $x = 4, y = 3$ *and* $z = 2$ and $\overline{F} = (2x^2 - 3z)\hat{\imath} - 2xy\hat{\jmath} - 4x\hat{k}$. (11)

(b) Find the inverse Laplace transform of (12)
(i) $\frac{3S+1}{(S-1)(S^2+1)}$
(ii) $\frac{1}{\sqrt{2S+3}}$

(c) State the convolution theorem. Use the convolution theorem to find $\mathcal{L}^{-1}\left\{\frac{1}{S^2(S+1)^2}\right\}$. (12)

4. (a) Using Heaviside expansion formula, find the inverse Laplace transform of (10)
$$\frac{S-1}{(S+3)(S^2+2S+2)}$$

(b) Using Laplace transform solve (15)
$y'' - 7y' + 2y = t + 4$, given that $y(0) = 0, y'(0) = -1$

(c) Evaluate $\int_c \overline{F}.d\overline{r}$ from $(0,0,0)$ to $(1,1,1)$ along the path C: the straight line joining $(0,0,0)$ to $(1,0,0)$; where (10)
$$\overline{F} = 2x\hat{\imath} - z\hat{\jmath} + x\hat{k}.$$

Section – B

5. (a) Define with example: (09)
   (i) Skew-symmetric matrix
   (ii) Diagonal matrix
   (iii) Orthogonal matrix

   (b) If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find A^{-1}. Also express matrix A and its inverse as products of elementary matrices performing for row and column elementary operations. Additionally verify that the A^{-1} you find is correct. (16)

   (c) Define linear independence of vectors. Check whether the vectors $\vec{a} = \hat{\imath} + 3\hat{\jmath} + 5\hat{k}$, $\vec{b} = 2\hat{\imath} + 5\hat{\jmath} + 9\hat{k}$ and $\vec{c} = -3\hat{\imath} + 9\hat{\jmath} + 3\hat{k}$ are linearly independent or not. If possible, express dependent vector as a linear combination independent vrctors. (10)

6. (a) What do you mean of the rank of a matrix? Reduce the matrix A, to its row echelon form, row canonical, and from normal form, where (18)

$$\begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$$

   Also determine its rank. Moreover, if possible, find A^{-1} as a by-product of the process to obtain its normal form.

   (b) What is the value of "K" for which the following system has: (i) an unique solution, (ii) many solution, and (iii) no solution? If possible, find the unique solution of the system using the Gauss elimination method (elementary row operations, where (17)

$$\begin{bmatrix} 2 & 3 & 12 \\ 0 & 6 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ k \end{pmatrix}$$

7. (a) Given $2x - y + 3z = 0$; $2x + 2y + z = 0$; $x - 4y + Tz = 0$. What is the value of 'T' so that the system of equations has: (i) trivial solution (ii) non-trivial solution? If possible final the non-trivial solution of the system of equations for same value of 'T'. (15)

   (b) Write down five properties/definitions for any eigensystem. Find the eigenvalues and corresponding eigenvectors of A, where $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Also find the eigenvalues and corresponding eigenvectors of the matrix $A - 4I$, where I is the identity matrix. (20)

8. (a) A space curve is defined by the position vector $cost\hat{\imath} + sint\,\hat{\jmath} + \sqrt{3}\hat{k}$. Find the unit tangent vector $\hat{T}$, normal vector $\hat{N}$, and binormal vector $\hat{B}$. Calculate the radius of curvature, acceleration, and to tangent and normal components of acceleration. Additionally, determine the trihedral plan that contains the vectors $\hat{T}$ and $\hat{N}$ at $t = \frac{\pi}{2}$. (28)

   (b) A force $\overline{F} = 10t\hat{\imath} + 15t^2\hat{\jmath}$ is acting on a moving particle with a mass of 5 units. Find its velocity such that the velocity at $t = 0$ is $5\hat{k}$. (07)

2022

MATH 2123 - Question Paper

3 supplied exam pages

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2022

MATH 2123

(Mathematics - III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any THREE questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) A space curve is given by the position vector $2\cos t\hat{i}+2\sin t\hat{j}+\hat{k}$. Find the unit tangent vector $\hat{T}$, unit normal vector $\hat{N}$, radius of curvature, and normal plane at $t=\frac{\pi}{2}$. Also, draw the space curve roughly. (28)

(b) How can you identify that a vector field is both solenoidal and rotational? Explain it and give a figure. (07)

2. (a) Find the value of a, b, c so that the directional derivative of $f=ayz^2+bzx+cx^2y^3$ at (2,-1,1) has a maximum of magnitude 64 in the direction parallel to y-axis. Hence find the directional derivative of $f=c$ along x-axis at the point (1,-1,2). (12)

(b) Define solenoidal vector and conservative vector field. Show that $\bar{A}=(2x^2+8xy^2z)\hat{i}+(3x^3y-3xy)\hat{j}-(4y^2z^2+2x^3z)\hat{k}$ is not solenoidal but $\bar{B}=xyz^2\bar{A}$ is solenoidal. (11)

(c) Define vector and scalar field with example. Find the work done in moving a particle in a force field given by $\bar{F}=3xy\hat{i}-5z\hat{j}+10x\hat{k}$ along the curve $x=t^2+1,\ y=2t^2,\ z=t^3;\ 1\le t\le 2$ (12)

3. (a) Evaluate $\iint_s \bar{F}.\hat{n}ds$, where s is the surface $2x+2y+z=4$ located in the first octant cutting by the planes x = 0, y = 0, z = 0. Given $\bar{F}=(3z)\hat{i}-2xy\hat{j}-4x\hat{k}$ (12)

(b) If $\bar{F}=(2x+y)\hat{i}+(3y-x)\hat{j}$, evaluate $\int_c \bar{F}.d\bar{r}$ where c is the curve in the xy plane consisting of the straight lines from (0,0) to (2,0) then to (3,2). (13)

(c) Evaluate $\iiint_v \nabla.\bar{F}dv$, where $\bar{F}=(2x^2-3z)\hat{i}-2xy\hat{j}-4x\hat{k}$ and v is the closed region bounded by the planes x = 1, y = 1, z = 1, & x = 3, y = 3, and z = 3. (10)

4. (a) Reduce the following matrix to its echelon form, canonical ...

Also find its rank. Here $A=\begin{bmatrix}0 & 1 & 0 & -2\\ -4 & 2 & 1 & 1\\ 2 & 0 & 1 & 0\end{bmatrix}$

(b) What is the value of k for which the following system of linear equations has (i) a unique solution, (ii) many solutions and (iii) no solution? If possible, find the unique solution for that particular value of k. $\begin{bmatrix}2 & 3 & k\\ 1 & 6 & 2k+3\\ 2 & 3 & 2\end{bmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$ (18)

Section – B

5. (a) Define Laplace transform. Find the Laplace transform of $t^3\sin t$ and hence, find the value of $\int_0^\infty t^3e^{-t}\sin t\,dt$ (12)

$\rightarrow L\{t^3\sin t\}$ at $s=1$ $\therefore$ hd → s=1 বসালে $\int_0^\infty t^3e^{-t}\sin t\,dt$ পাবো

(b) Let $F(t)=\begin{cases}3t & ;0<t<2\\ 6 & ;2<t<4\end{cases}$ (12)

hd

Where F(t) has period 4. Find L{F(t)}.

(c) State convolution theorem. Apply this theorem to find $L^{-1}\left\{\frac{1}{S^2(S+1)^2}\right\}$

$\rightarrow f(s)=\frac{1}{s^2}$ $L^{-1}\{f(s)\}=t$

$g(s)=\frac{1}{(s+1)^2}$, $L^{-1}\{g(s)\}=te^{-t}$

h.d

$\frac{A}{S}+\frac{B}{S-1}+\frac{CS+D}{S^2+4}$

$=A(S-1)(S^2+4)+BS(S^2+4)+(CS+D)(S^2-S)$

6. (a) Find the inverse Laplace transform of $\dfrac{S+4}{S(S-1)(S^2+4)}$ (12)

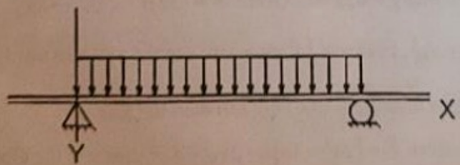
(b) Solve the Differential equation $y''(t)+4y'(t)+4y(t)=4e^{-2t}$, $y(0)=-1$, $y'(0)=4$, with the help of Laplace transform. (12)

(c) Using Laplace transform solve the integral equation $F(t)=1+2\int_0^t e^{-2x}F(t-x)dx$ (11)

$f(t)=e$

Term-20 5(a)

7. (a) A beam is hinged at its ends x = 0 and x = l, which carries a uniform load w_o per unit length as shown in the figure below. Find the deflection at any point. (14)



(b) Define orthogonal matrix. If $(l_r, m_r, n_r), r=1,2,3$ be the direction cosine of three mutually perpendicular lines referred to an orthogonal coordinates system then prove that $A=\begin{bmatrix}l_1 & m_1 & n_1\\ l_2 & m_2 & n_2\\ l_3 & m_3 & n_3\end{bmatrix}$ is orthogonal. (11)

(c) Find the eigen values and the eigen vector corresponding to the largest eigen value of A, $A=\begin{bmatrix}0.5 & 0.5\\ 0.5 & 0.5\end{bmatrix}$ (10)

8. (a) Given $2x - y + 3z = 0$, $3x + 2y + z = 0$, $x + 3y - 2z = 0$. If A is the coefficient matrix and x will be the solution of the above system of linear equations then find the solution. Again find the solution of the following system for $(BA)X = 0$, where the (20)

transition matrix $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

Is the solution space changed? Explain.

(b) (15)

Find A^{-1} where $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$. Given Ax = H

Where $H = [1 \quad 0 \quad 1]'$ then find the value of x.

2021

MATH 2123 - Question Paper

2 supplied exam pages

... University of Engineering & Technology

Department of Building Engineering & Construction Management

B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2021

Math 2123

(Mathematics-III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) A particle moves along the path $r = 2\cos t\,\hat{\imath} + 2\sin t\,\hat{\jmath} + \hat{k}$ (35)
   (i) Find velocity, tangent vector, Normal vector and Binormal vector at t = 1.
   (ii) Find radius of curvature and torsion at t = 0
   (iii) Find osculating plane at t = 1.
   (iv) Also sketch the path and comment about it.

2. (a) Classify the vector field $\frac{\bar{r}}{r^2}$ by using differential operator '∇'. (10)

   (b) Find the angle of the two surfaces $x^2 + y^2 + z^2 = 12$ and $xyz = 8$ at (2, 2, 2). Also find the tangential plane to the surface $x^2 + y^2 + z^2 = 12$ at that point. Also find the directional derivatives of the scalar point function $f(x,y,z) = xyz - 8$ along x axis at that point. (15)

   (c) Find the workdone in a force field given by $\bar{F} = 3xy\hat{\imath} - 5z\hat{\jmath} + 10x\hat{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 0$ to $t = 3$. (10)

3. (a) Evaluate $\iint_s \bar{q}.\hat{n}dS$ where S is a closed surface bounded by the cylinder $x^2 + y^2 = 9$, $z = 0$ to $z = 2$, $x = 0$, $y = 0$ and $\bar{q} = 4yx^2\hat{\imath} + 2y^2z\hat{\jmath} - z^2\hat{k}$. (15)

   (b) Find the volume of the region bounded by the planes $2x + 3y + z = 1$, $x = 0$, $y = 0$, $z = 0$. (10)

   (c) Test linear dependent and independent of vector $\bar{r}_1 = 3\hat{\imath} + \hat{\jmath} - 2\hat{k}$, $\bar{r}_2 = \hat{\imath} + 3\hat{\jmath} + \hat{k}$ and $\bar{r}_3 = \hat{\imath} - 5\hat{\jmath}$. Also, if possible, express the dependent vector as a linear combination of independent vectors. (10)

4. (a) Define with examples: (i) Skew – symmetric matrix (ii) Singular matrix (iii) Diagonal matrix. (12)

   (b) Find the matrix A and the matrix B, such that $3A - 2B = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$ and $-4A + B = \begin{bmatrix} -1 & 2 \\ -4 & 3 \end{bmatrix}$. (10)

   (c) When does a matrix have its inverse? If possible find the inverse of $\begin{bmatrix} 2 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$ (13)

Section – B

5\. (a) Define the Laplace transform. Find the Laplace transform of the following functions: (14)

(i) $F(t) = \begin{cases} 3t, & 0 < t < 2 \\ 6, & 2 < t < 4 \end{cases}$, Where F(t) has period 4.

(ii) $F(t) = 3t^4 + 4e^{-3t} + t^2 \cos 2t$

(b) Find the Laplace transform of $\sin\sqrt{t}$. Also find $L\left\{\frac{\cos\sqrt{t}}{\sqrt{t}}\right\}$. (11)

(c) If $F(t) = 5\sin 3\left(t - \frac{\pi}{4}\right)$ for $t > \frac{\pi}{4}$ and 0 for $t < \frac{\pi}{4}$, find $L\{F(t)\}$. (10)

6\. (a) Find the inverse Laplace transform of $\frac{s+5}{(s+1)(s^2+4)}$. (12)

(b) Define the inverse Laplace transform. Find $L^{-1}\left\{\frac{3s+1}{(s-1)(s^2+1)}\right\}$ by using the Heaviside expansion formula. (12)

(c) Find the inverse Laplace transform of (i) $\frac{s^2+3}{s(s^2+9)}$ (ii) $\log\frac{s^2-1}{s^2}$ (11)

7\. (a) Using Convolution Theorem find $L^{-1}\left\{\frac{s^2}{(s^2+1)(s^2+2)}\right\}$ (11)

(b) Solve the following differential equation by using the Laplace transform: (14)
$y'' - 3y' + 2y = e^{2t}$; $y(0) = -3$, $y'(0) = 5$.

(c) Find the eigen values and eigen vectors for the matrix, A = $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ (10)

8\. (a) Solve the system of equations, (18)

$$x_1 - 2x_2 + 3x_3 - 4x_4 = 5$$
$$x_1 - 3x_2 + 7x_3 - 6x_4 = 7$$
$$2x_1 - 5x_2 + 10x_3 - 11x_4 = 12$$

(b) Find all non-trivial solutions of (17)

$$3x_1 + 5x_2 - 2x_3 = 0$$
$$x_1 - 3x_2 + 4x_3 = 0$$
$$2x_1 + x_2 + x_3 = 0$$

$\frac{s}{s+1}\cdot\frac{s}{s+2}$

$\cos t \cdot \cos\sqrt{2}t$

$\int_0^t \cos\sqrt{2}v \cos(t-v)\,dv$

$= \frac{1}{2}\int_0^t \cos(\sqrt{2}v - v + t) + \cos(\sqrt{2}v + v - t)\,dv$

$= \frac{1}{2}\left[\frac{\sin(\sqrt{2}v - v + t)}{\sqrt{2}-1} + \frac{\sin(\sqrt{2}v + v - t)}{\sqrt{2}+1}\right]_0^t$

$= \frac{1}{2}\left(\frac{\sin\sqrt{2}t}{\sqrt{2}-1} + \frac{\sin\sqrt{2}t}{\sqrt{2}+1} - \frac{\sin t}{\sqrt{2}-1} + \frac{\sin t}{\sqrt{2}+1}\right)$

$= \frac{1}{2}(2\sqrt{2}\sin\sqrt{2}t - 2\sin t)$

$= \sqrt{2}\sin\sqrt{2}t - \sin t$

$\begin{vmatrix} 3 & 5 & -2 \\ 1 & -3 & 4 \\ 2 & 1 & 1 \end{vmatrix}$

$= 3(-7) - 5(-7) - 2(7)$

$= 0$

So, trivial value came

$\begin{bmatrix} 3 & 5 & -2 \\ 1 & -3 & 4 \\ 2 & 1 & 1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$R_2' \to 3R_2 - R_1$, $R_3' \to 3R_3 - 2R_1$ $\begin{bmatrix} 3 & 5 & -2 \\ 0 & -14 & 14 \\ 0 & -7 & 7 \end{bmatrix}$

$R_3' \to 2R_3 - R_2$ $\begin{bmatrix} 3 & 5 & -2 \\ 0 & -14 & 14 \\ 0 & 0 & 0 \end{bmatrix}$

$3x_1 + 5x_2 - 2x_3 = 0$ $x_3 = a$

$-x_2 + x_3 = 0$

$x_2 = a$

$x_1 = -a$

Soln

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = a\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

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2020

MATH 2123 - Question Paper

2 supplied exam pages

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management

B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2020

Math 2123

(Mathematics III)

Full Marks: 120 Time: 1.5 hrs

N.B.
i) Answer any two questions from each section in separate script.
ii) Figures in the right margin indicate full marks
iii) Assume reasonable values for any missing data.

Section – A

1. (a) Find the value of λ so that the vectors $\lambda\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$ and $3\hat{i} - 4\hat{j} + 5\hat{k}$ are not coplanar. (10)

   (b) Find $\vec{a} \times (\vec{b} \times \vec{c})$ and $(\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c}$ where $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + \hat{j} - \hat{k}$, $\vec{c} = \hat{i} - \hat{j} + \hat{k}$. Also find the angle between $\vec{a} \times (\vec{b} \times \vec{c})$ and $\vec{b} \times \vec{c}$. (20)

2. (a) The derivative of $f(x,y)$ at $P(1,2)$ in the direction of $\hat{i} + \hat{j}$ is $2\sqrt{2}$ and in the direction of $-2\hat{j}$ is -3. What is the derivative of $f(x,y)$ in the direction of $-\hat{i} - 2\hat{j}$? Give reasons for your answer. (18)

   (b) Find the work done in a moving particle in the force field, $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the curve defined by $x^2 = 4y$, $3x^3 = 8z$ from $x = 0$ to $x = 2$. (12)

3. (a) Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ then find all eigen values and corresponding eigen vectors. (10)

   (b) The augmented matrix of a linear system has the form (14)
   $$\begin{bmatrix} a & 1 & \vdots & 1 \\ 2 & (a-1) & \vdots & 1 \end{bmatrix}$$
   (i) Determine the values of a for which the linear system is consistent.
   (ii) When it is consistent, does the system have a unique solution or infinitely many solutions?

   (c) Find the directional derivatives of $\phi = x^2yz + 4xz^2$ at $(1,-2,-1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$. (06)

Section – B

4. (a) Define Laplace transform. Find $L\{f(t)\}$ where, (10)
   $$f(t) = \begin{cases} \sin t ; & 0 \le t < \pi/4 \\ \sin t + \cos(t - \pi/4) ; & t \ge \pi/4 \end{cases}$$
   same

   $f(t) = \sin t + \begin{cases} 0 & 0 \le t < \pi/4 \\ \cos(t - \frac{\pi}{4}) & t \ge \pi/4 \end{cases}$

   $= \sin t + \cos(t - \frac{\pi}{4})\, u(t - \frac{\pi}{4})$

   $= \frac{1}{s^2+1} + e^{-\frac{\pi}{4}s} \frac{s}{s^2+1}$

   $= \frac{1 + s e^{-\frac{\pi s}{4}}}{s^2+1}$

   (b) Find the Inverse Laplace transform, $L^{-1}\left\{\frac{1}{(S+2)^2(S-2)}\right\}$ (10)

   partial fraction

   distinct

   $Q(s) = (s+2)(s^2-4) = s^3 + 2s^2 - 4s - 8$

   (c) Find the value of Integral, $\int_0^\infty t^3 e^{-t} \sin t\, dt$ (10)

$L^{-1}\{f(s)\} = F(t)$

$L^{-1}\{g(s)\} = G(t)$

$L^{-1}\{f(s)\,g(s)\} = \int_0^t F(u)\,G(t-u)\,du = F(t) * G(t)$

5. (a) Using Laplace transform solve the following integral equation, (15)

$$F(t) = 1 + 2\int_0^t F(t-x)e^{-2x}dx$$

(b) Using Laplace transform solve the differential equation: (15)

$$Y^{(iv)}(t) - 16y(t) = 30sint$$

subject to $y(0) = 0, y'(0) = 2, y''(\pi) = 0, y'''(\pi) = -18.$

6. (a) Define adjoint of a matrix. Find the inverse of (15)

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 5 & 7 & 8 \\ 4 & 10 & 12 \end{bmatrix}$$

(b) Find the rank of the following real matrix: (15)

$$\begin{bmatrix} a & 1 & 2 \\ 1 & 1 & 1 \\ -1 & 1 & 1-a \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ a & 1 & 2 \\ -1 & 1 & 1-a \end{bmatrix}$$

Where 'a' is a real number.

$(1-a)(2-a)$

$2-3a+a^2-4+2a$

$R_2' \to R_2 - aR_1$, $R_3' \to R_3 + R_1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1-a & 2-a \\ 0 & 2 & 2-a \end{bmatrix}$$

$R_3' \to (1-a)R_3 - 2R_2$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1-a & 2-a \\ 0 & 0 & a^2-a-2 \end{bmatrix}$$

Rank $= 3$ | $a = 2, -1$

if $a \neq 1, 2$ ∴ rank $= 3$

$K = 2$, rank 2

or, $K = -1$ " 2

$(k-2)(k+1)$

$k^2 - k - 2$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1-a & 2-a \\ 0 & 0 & (a-2)(a+1) \end{bmatrix}$$

rank $= 3$ if $a \neq 2, -1$

rank $= 2$ if $a = 2, -1$

$\int_0^t f(u)\,g(t-u)\,du$

$F(t) = 1 + 2\int_0^t F(t-x)\,e^{-2x}\,dx$

$F(t) = 1 + 2[F(t) * e^{-2t}]$

$f(s) = \frac{1}{s} + 2\left[f(s) \cdot \frac{1}{s+2}\right]$

laplace trns $f(s) = \frac{1}{s} + \frac{2f(s)}{s+2}$

$(s+2)f(s) = \frac{s+2}{s} + 2f(s)$

$f(s) = \frac{1}{s} + \frac{2}{s^2}$

inverse laplace $F(t) = 1 + 2t$

2019

MATH 2123 - Question Paper

3 supplied exam pages

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2019
Math 2123
(Mathematics-III)

Full Marks: 210 Time: 3 hrs.

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define homogeneous system of linear equations. Find a cubic polynomial $P(x) = a + bx + cx^2 + dx^3$ such that, $P(1) = 1, P'(1) = 5$, $P(-1) = 3$ and $P'(-1) = 1$. (13)

(b) What is meant by orthogonal matrix? Determine whether the matrix A is orthogonal or not. (10)

$$A = \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ 2/3 & -1/3 & 2/3 \\ -1/3 & 2/3 & 2/3 \end{bmatrix}$$

(c) Determine the values of 'a' for which the following system has no solutions, exactly one solution or infinitely many solutions. (12)

$$x + 2y + z = 2$$
$$2x - 2y + 3z = 1$$
$$x + 2y + (a^2 - 3) = a$$

2. (a) Define rank of the matrix. Find the rank of the matrix B, where k is a real number. (10)

$$B = \begin{bmatrix} k & 1 & 2 \\ 1 & 1 & 1 \\ -1 & 1 & 1-k \end{bmatrix}$$ hd at 20

(b) Using elementary row operations, find the inverse of the matrix: (12)

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

(c) If $M = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ then find all eigenvalues of M and the corresponding eigenvectors. Also evaluate an invertible matrix P such that $P^{-1}MP$ is diagonal. (13)

3. (a) What is symmetric matrix? Find all values of a, b and c for which matrix A is symmetric. (10)

$$A = \begin{bmatrix} 2 & a - 2b + 2c & 2a + b + c \\ 3 & 5 & a + c \\ 0 & -2 & 7 \end{bmatrix}$$

(b) Define Laplace transform. Find Laplace transform of (12)

$$t^2 e^{-t} \cos t + (t-1)^2$$

(c) Define unit step function. Express the following function in terms of unit step. Hence find its Laplace transform. (13)

$$f(t) = \begin{cases} 2 & 0 \le t < 3 \\ 2t & 3 \le t < 5 \\ \frac{1}{7}t^2 & t \ge 5 \end{cases}$$

4. (a) State convolution theorem for Inverse Laplace transform. Using partial fraction decompositions evaluate (12)

$$L^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\}$$

$5s+3 = A(s^2+2s+5) + (Bs+C)(s-1)$

$A = 3/4$, $B = -3/4$, $C = 3/4$

$= \frac{3}{4}e^{t} - \frac{3}{4}e^{-t}\cos 2t$

(b) Using the concept of Laplace transform prove that

$$\int_0^\infty \frac{\sin^2 t}{t^2}\, dt = \frac{\pi}{2}$$

(c) Graph of a function f(t) is given below. Find Laplace transform of f(t). (12)

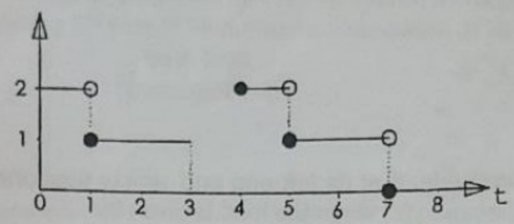

Figure – Question 4 (c)

Section – B

5. (a) The scalar triple product can be interpreted as the volume of a parallelepiped. Now you give the details about this interpretation and then find the volume of a paralleped determined by the vectors (12)

directional $2\hat{i} - 3\hat{j} + 4\hat{k},\ \hat{i} + 2\hat{j} - \hat{k},\ 3\hat{i} - \hat{j} + 2\hat{k}$

(b) Find the derivative of the function $f(x,y,z) = yz + zx + xy$ at the point (1, -1, 2) in the direction of $\vec{A} = 3\hat{i} + 6\hat{j} - 2\hat{k}$. (10)

(c) Show that $\vec{F} = (e^x \cos y + yz)\,\hat{i} + (xz - e^x \sin y)\,\hat{j} + (xy + z)\,\hat{k}$ is conservative and find a potential function for it. (13)

6. (a) Show that $div\,(\varphi \vec{A}) = (grad\,\varphi) . \vec{A} + \varphi\, div\, \vec{A}$ and verify for $\varphi = e^{xyz}$ and $\vec{A} = ax\,\hat{i} - by\,\hat{j} + cz\,\hat{k}$. (10)

(b) If $\vec{A} = (y^2 + 2x)\,\hat{i} + (3y - 4x)\,\hat{j}$ evaluate $\oint_c \vec{A} \cdot d\vec{r}$ around the triangle c of the following figure in the indicated direction. (11)

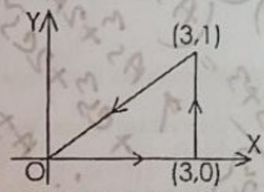

Figure: Question – 6 (b)

(c) Evaluate (14)

$$\iint_s (\vec\nabla \times \vec{F}) . \hat{n}\, ds$$

where $\vec{F} = (x + 2y)\,\hat{i} - 3z\,\hat{j} + x\,\hat{k}$ and s is the surface of $2x + y + 2z = 6$ located in the first octant.

7. (a) Evaluate (15)

$$\iiint_v (\vec\nabla \times \vec{F})\, dv$$

where v is the closed region bounded by the planes $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$ we have $F = xyz\,i + yz\,j + y^2 k$

(b) Find the actual (acute) angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point (2, -1, 2). (10)

(c) For $\vec{A} = 2yz\,\hat{i} - x^2 y\,\hat{j} + xz^2\,\hat{k}$ and $\varphi = 2x^2yz^3$ find $(\vec{A} \times \vec\nabla)\,\varphi$ and $\vec{A} \times (\vec\nabla\varphi)$. (10)

3. (a) A beam is embedded at its left end and simply supported at its right end. Find the deflection y(x) when the load is given by (20)

$$w(x) = \begin{cases} w_0; 0 < x < \frac{L}{2} \\ 0; \frac{L}{2} \le x < L \end{cases}$$

(b) Using Laplace transform solve the differential equation (15)

$$y'' + 2y' + 5y = e^{-t}\sin t$$
$$y(0) = 0 \quad y'(0) = 1$$

b $L\{y\} = Y$

$\therefore s^2Y - s.0 - 1 + 2(sY - 0) + 5Y = \frac{1}{(s+1)^2+1}$

$\Rightarrow Y(s^2+2s+5) = \frac{1}{s^2+2s+2} + 1$

$Y = \frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}$

$\therefore s^2+2s+3 = (As+B)(s^2+2s+5) + (Cs+D)(s^2+2s+2)$

$\Rightarrow As^3+2As^2+5As+Bs^2+2Bs+5B + Cs^3+2Cs^2+2Cs+Ds^2+2Ds+2D$

$\therefore A+C=0$, $A=-C$

$(2A+B+2C+D)=1$

$5A+2B+2C+2D=2$

$5B+2D=3$

$A=0$, $B=\frac{1}{3}$, $C=0$, $D=\frac{2}{3}$

$Y = \frac{s^2+2s+3}{\{(s+1)^2+1\}\{(s+1)^2+2^2\}}$

$= \frac{(s+1)^2+2}{\{(s+1)^2+1\}\{(s+1)^2+4\}}$

$= \frac{x+2}{(x+1)(x+4)}$ $\quad (s+1)^2 = x$

$= \frac{1}{(x+1)3} + \frac{-2}{(x+4)(-3)}$

$Y = \frac{\frac{1}{3}}{(s+1)^2+1} + \frac{\frac{2}{3}}{(s+1)^2+2^2}$

$\therefore y = \frac{e^{-t}}{3}\sin t + \frac{e^{-t}}{3}\sin 2t$

$= \frac{e^{-t}}{3}(\sin t + \sin 2t)$

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2018

MATH 2123 - Question Paper

2 supplied exam pages

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2018
Math 2123
(Mathematics III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define orthogonal matrix, Hermitian matrix, Idempotent matrix and lower triangular matrix with example. (12)

(b) Express the matrix $A=\begin{bmatrix}1 & 1 & 3\\ 5 & 2 & 6\\ 2 & -1 & -3\end{bmatrix}$ as the sum of a symmetric and skew-symmetric matrix. (10)

(c) Find the inverse of the following matrix applying elementary row transformation. (13)

$A=\begin{bmatrix}-2 & 1 & 3\\ 0 & -1 & 1\\ 1 & 2 & 0\end{bmatrix}$

2. (a) Reduce the matrix $A=\begin{bmatrix}2 & 1 & -1 & 3\\ -3 & 0 & 1 & -2\\ -4 & 5 & 0 & -1\end{bmatrix}$ to echelon form and then to canonical form and then to normal form. Hence find its rank. (12)

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(b) Find the non-singular matrices P and Q such that PAQ is the normal form, where $A=\begin{bmatrix}2 & -1 & 3 & 4\\ -1 & -3 & 0 & 1\\ 1 & -2 & -1 & 2\end{bmatrix}$. (15)

(c) Show that the diagonal elements of a skew-symmetric matrix are zero. (08)

3. (a) When the system of linear equations are called consistent? Determine the value of a so that the following equations has (i) no solution (ii) non-zero unique solution (iii) Infinite solution: (14)

$x+y-z=1$; $2x+3y+az=3$; $x+ay+3z=2$. After taking

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(b) Check the following system of linear equations are consistent or not. If consistent then find the non-trivial solutions of them: (11)

$2x_1-3x_2+x_3+5x_4=2$
$-x_1-2x_3+x_4=3$
$3x_1+2x_2-x_3+2x_4=-1$

Non trivial only for homogeneous

(c) If $A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\end{bmatrix}$, then show that trace (AA')=trace (A'A). (10)

Just Calculation

4. (a) Define elementary matrix and equivalent matrix with examples. (07)

(b) Show that if A is involuntary then $\frac{1}{2}(I+A)$ and then $\frac{1}{2}(I-A)$ are idempotent. (08)

A is involuntary so, $A^2=I$

$B=\frac{1}{2}(I+A)$ $\quad C=\frac{1}{2}(I-A)$

$B^2=\frac{1}{4}(I^2+2A\cdot I+A^2)$ $\quad C^2=\frac{1}{4}(I-2A+I)$

$=\frac{1}{4}(2I+2A)$ $\quad =\frac{1}{2}(I-A)$

$=\frac{1}{2}(I+A)$ $\quad =C$

$=B$

$\therefore$ B, C both idempotent

Page 1 of 2

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(c) Define Eigen values and Eigen vector and write its three properties. And also find the Eigen values and Eigen vector corresponding to one of the Eigen values of the matrix $B=\begin{bmatrix}1 & -3 & 3\\ 3 & -5 & 3\\ 6 & -6 & 4\end{bmatrix}$

Section – B

5. (a) State the sufficient conditions for the existence of Laplace transform of functions. Find Laplace transform of $f(t) = \frac{1}{\sqrt{t}}$; $0<t<\infty$ if exist. (12)

(b) Find the Laplace transform of g(t), where $g(t)=\int_0^t \frac{\sin u}{u}du + te^{2t}\cos 2t$. (13)

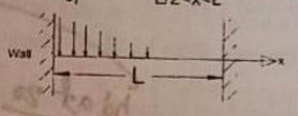
(c) Using convolution theorem find the inverse Laplace transform of $\frac{1}{s^2(s+1)^2}$. (10)

6. (a) Using Laplace transform solve the following equation (15)

$$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = e^t \text{ with } x = 2, \frac{dx}{dt} = -1 \text{ at } t = 0$$

(b) A beam of length L is embedded at both ends as shown in figure below. Find the deflection of the beam when the load is given by (20)
$w(x)= w_0(1-\frac{2}{L}x)$; $0<x<\frac{L}{2}$

0; L/2<x<L



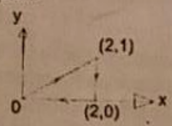
7. (a) Geometrically interpret the scalar triple product and use it to determine whether the following set of vectors are linearly dependent or not: (10)
$\vec{A}=\hat{i} - 3\hat{j} + 2\hat{k}, \vec{B} = 3\hat{i} + 2\hat{j} - \hat{k}, \vec{C} = 2\hat{i} + \hat{j} - 3\hat{k}$.

(b) A force $\vec{F} = 3\hat{i} + 2\hat{j} - 4\hat{k}$ is applied at the point (1,-1, 2). Find the moment of the force about the point (2, -1, 3). (10)

(c) Show that the vector field represented by (15)
$\vec{F} = (z^2 + 2x + 3y)\hat{i} + (3x + 2y + z)\hat{j} + (y + 2zx)\hat{k}$ is not solenoidal. Is $\vec{F}$ conservative? If so, find its scalar potential ø.

8. (a) Find the value of a and b so that the surface ax^2-byz= (a+z) will be orthogonal to the surface $4x^2y+z^3=4$ at the point (1, -1, 2). (10)

(b) If $\vec{F} = (2x + y^2)\hat{i} + (3y - 4x)\hat{j}$ evaluate the $\int_c \vec{F}.d\vec{r}$ around the triangle c of below figure in the directed path: (10)



(c) Evaluate $\iint_s \underline{A}.\underline{n}\,ds$, where $\underline{A} = z\underline{i} + x\underline{j} - 3y^2z\underline{k}$ and s is the surface of the cylinder $x^2+y^2=16$ included in the first octant between z=0 and z=5. (15)

2017

MATH 2123 - Question Paper

2 supplied exam pages

Khulna University of Engineering & Technology

Department of Building Engineering and Construction Management

B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2017

Math 2123

(Mathematics-III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define integral transform and kernel of transform. Also find the kernel of Laplace transform. (10)

(b) Find the Laplace transform of $te^{2t}\sin 3t$ (10)

(c) Solve the following I.V. P. by Laplace transform (15)

$$Y'' + 9Y = \cos 2t, \quad Y(0) = 1, \quad Y(\pi/2) = -1$$

2. (a) Find the inverse Laplace transform of $\dfrac{11S^2-2S+5}{(S-2)(2S-1)(S+1)}$ Using Heaviside expansion formula. (10)

(b) A cantilever beam is clamped at the end $x = 0$ and is free at the end $x = l$. It carries a load per unit length given by (15)

$$W(x) = \begin{cases} w_0, & 0 < x < l/2 \\ 0, & l/2 < x < l. \end{cases}$$

Find the deflection. Using Laplace transform

(c) In each determine whether the vectors are linearly independent or linearly dependent (10)

$\underline{A} = 2\underline{i} + \underline{j} - 3\underline{k}$, $\underline{B} = \underline{i} - 4\underline{k}$, $\underline{C} = 4\underline{i} + 3\underline{j} - \underline{k}$

$\underline{A} = \underline{i} - 3\underline{j} + 2\underline{k}$, $\underline{B} = 2\underline{i} - 4\underline{j} - \underline{k}$, $\underline{C} = 3\underline{i} + 2\underline{j} - \underline{k}$

3. (a) If $\underline{A} = \dfrac{\underline{r}}{|\underline{r}|}$ find grade div $\underline{A}$ (08)

(b) Find an equation for the tangent plane to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$ (12)

(c) A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$, $z = 2\sin 3t$, where t is the time. Find the magnitudes of the velocity and acceleration at time, $t = 0$. (15)

4. (a) Test the vector $\underline{A} = (6xy + z^3)\underline{i} + (3x^2 - z)\underline{j} + (3xz^2 - 4)\underline{k}$, is irrotational or not. If the vector is irrotational find its scalar potential. (13)

(b) Find the Directional derivative of $P = 4e^{2x-y+z}$ at the point $(1,1,-1)$ in a direction toward the point $(-3,5,6)$. (10)

(c) If $\underline{A} = (2y+3)\underline{i} + xz\underline{j} + (yz - x)\underline{k}$, evaluate $\int_c \underline{A}\cdot d\underline{r}$ along the following path C: the straight lines from $(0,0,0)$ to $(0,0,1)$ then to $(0,1,1)$, and then to $(2,1,1)$. (12)

Section – B

5 (a) Define with example (i) skew- symmetric matrix, (ii) principal sub matr.., elementary matrix, (iv) diagonal matrix.

(b) Express the matrix A in the form P+Q, where P is a symmetric matrix and Q is a skew- symmetric matrix;

$$A = \begin{bmatrix} 3 & 3 & -1 \\ 0 & 3 & -2 \\ -1 & 2 & 2 \end{bmatrix}$$

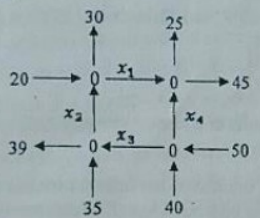
(c) Reduce the matrix to its echelon form then to its canonical form then to its normal form. And also find its rank where (15)

$$A = \begin{bmatrix} 2 & 1 & 5 & 1 & 5 \\ 1 & 1 & -3 & -4 & -1 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{bmatrix}$$

6 (a) Find the inverse of the matrix $A = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -3 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$ using elementary transformation. (15)

(b) What do you mean by a consistent system of linear equations? What are the different method of solving non- homogenous system of linear equations? Investigate for consistency of the following equations and if possible find the solutions of $4x - 2y + 6z = 8$, $x + y - 3z = -1$, $15x - 3y + 9z = 21$ by using matrix. (20)

7 (a) Find the traffic flow in the adjoining network of one way streets. Is the solution unique? (13)



(b) Define eigen value and eigen vector of a matrix. Find all eigen values and associated eigen vectors of the following matrix: $A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix}$ (22

8 (a) Find adj A and hence A^{-1}, where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ (1

(b) Verify the Cayley- Hamilton theorem for the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ (1

(c) Find the nonsingular matrices P & Q such that PAQ is normal form where, (1

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

2016

MATH 2123 - Question Paper

3 supplied exam pages

Khulna University of Engineering & Technology

Department of Building Engineering & Construction Management

B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2016

Math 2123

(Mathematics - III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define with example: (i) Matrix, (ii) Scalar matrix, (iii) Hermitian matrix and (iv) Trace of a matrix. (12)

(b) If A is a square matrix, express the matrix A as a sum of a Hermitian matrix and a skew-Hermitian matrix. (10)

(c) If $A=\begin{bmatrix} 1 & 0 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$ (13)

and I is the unit matrix of order 3, evaluate $A^3+3A^2-7A+9I$.

2. (a) Define inverse of a matrix. When does it exist? Write down the following system of equations in matrix form and find the inverse of coefficient matrix by using elementary (row) transformations. Hence solve the system using matrix method: (17)

$$\begin{aligned} x-3y-8z &= -10 \\ 3x+y \qquad &= -4 \\ 2x+5y+6z &= 13 \end{aligned}$$

(b) Discuss different types of elementary transformations which are valid on a matrix. (05)

(c) Find the row-reduced echelon form and hence the normal form of the matrix (13)

$A=\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$. What is its rank?

3. (a) For what value (s) of λ, the system of equations, (15)

$$\begin{aligned} 3x-y+4z &= 3 \\ x+2y-3z &= -2 \\ 6x+5y+\lambda z &= -3 \end{aligned}$$

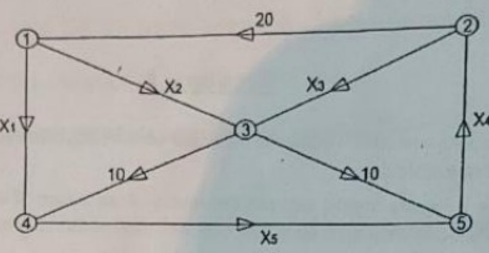
Possesses (i) a unique solution? and (ii) infinitely many solutions? Determine the solution for case (ii).

(b) Why a homogenous system is always consistent? What do we mean by trivial and non-trivial solution of a homogeneous system? Solve the system: (15)

$$\begin{aligned} 4x_1-x_2+2x_3+x_4 &= 0 \\ 2x_1+3x_2-x_3-2x_4 &= 0 \\ 7x_2-4x_3-5x_1 &= 0 \\ 2x_1-11x_2+7x_3+8x_4 &= 0 \end{aligned}$$

(c) Define orthogonal matrix. Is the matrix $\frac{1}{3}\begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}$ orthogonal?

4. (a) Set up a system of linear equations to represent the network shown in the figure: (12)



What type of solution does the system possess?

(b) Define eigen values and eigen vectors. Determine the Eigen values and the corresponding eigen vectors of the matrix. (18)

$$A=\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}.$$

(c) If a non-singular matrix A is symmetric show that A^{-1} is also symmetric. (05)

Section B

5. (a) Prove that $\underline{A}\times(\underline{B}\times\underline{C})=\underline{B}(\underline{A}.\underline{C})-\underline{C}(\underline{A}.\underline{B})$ for any three vectors $\underline{A}$, $\underline{B}$ and $\underline{C}$. Using this result, show that $\underline{A}\times(\underline{B}\times\underline{C})+\underline{B}\times(\underline{C}\times\underline{A})+\underline{C}\times(\underline{A}\times\underline{B})=0$. (12)

(b) The diagonal of a parallelogram are given by the vectors $\underline{A}=3\hat{i}-4\hat{j}-\hat{k}$ and $\underline{B}=2\hat{i}+3\hat{j}-6\hat{k}$. Prove that the parallelogram is a rhombus and determine the length of its sides. (12)

(c) Determine whether the vectors are linearly independent or linearly dependent: $\underline{A}=2\underline{i}+\underline{j}-3\underline{k}$, $\underline{B}=\underline{i}-4\underline{k}$, $\underline{C}=4\underline{i}+3\underline{j}-\underline{k}$. (11)

6. (a) Find equations for the tangent plane and normal line to the surface $z=x^2+y^2$ at the point $(2,-1,5)$. (12)

(b) In what direction from the point $(2,1,-1)$ in the directional derivative of $\varphi=x^2yz^3$ a maximum? What is the magnitude of this maximum? (12)

(c) Evaluate (i) $\nabla^2(\ln r)$ (ii) $\underline{\nabla}f(r)$. (11)

7. (a) The acceleration $\underline{a}$ of an object at any time t is given by $\underline{a}=-g\hat{j}$, where g is a constant. At $t=0$ the velocity is given by $\underline{V}=V_0\cos\theta_0\hat{i}+V_0\sin\theta_0\hat{j}$ and the displacement $\underline{r}=\underline{0}$. Find $\underline{V}$ and $\underline{r}$ at any time $t>0$. (12)

(b) If $\underline{A} = (2y+3)\hat{i} + xz\hat{j} + (yz-x)\hat{k}$, evaluate $\int_c \underline{A}.d\underline{r}$ along the following paths C.

(i) the straight lines from $(0,0,0)$ to $(0,0,1)$, then to $(0,1,1)$ and then to $(2,1,1)$.

(ii) the straight line joining $(0,0,0)$ and $(2,1,1)$.

(c) Evaluate $\iiint_V (2x+y)dV$, where V is the closed region bounded by the cylinder $z = 4-x^2$ and the planes $x=0$, $y=0$, $y=2$ and $z=0$. (11)

8. (a) Find Laplace transforms of (i) $F'(t)$ (ii) $e^{4t} + 4t^3 - 2\sin 3t + 4\cos 5t$. (12)

(b) Evaluate $L^{-1}\{\frac{1}{s^3(s^2+4)}\}$. (11)

(c) Solve the differential equation using the Laplace transform: $Y'' + a^2 Y = F(t)$; $Y(0)=1$, $Y'(0)=-2$. (12)

$L^{-1}\frac{1}{s^3(s^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{Ds+E}{s^2+4}$

$1 = A(s^4+4s^2) + B(s^3+4s) + C(s^2+4) + (Ds+E)s^3$

$= As^4 + 4As^2 + Bs^3 + 4Bs + Cs^2 + 4C + Ds^4 + Es^3$

$A + D = 0$ — (i)

$B + E = 0$ — (ii)

$4A + C = 0$ — (iii)

$4B = 0 \Rightarrow B = 0$

$4C = 1 \quad C = 1/4$

$\therefore E = 0$

$4A = 1/4 \quad D = \frac{1}{16}$

$A = -1/16$

$\therefore \frac{-1/16}{s} + \frac{1/4}{s^3} + \frac{1/16\, s}{s^2+4}$

$= -\frac{1}{16} + \frac{1}{4}\frac{t^2}{2!} + \frac{1}{16}\cos 2t$

$= -\frac{1}{16} + \frac{1}{8}t^2 + \frac{1}{16}\cos 2t$

2015

MATH 2123 - Question Paper

2 supplied exam pages

Khulna University of Engineering & Technology
Department of Building Engineering & Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2015
Math 2123
(Mathematics - III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define with example (any three): (12)
(i) Equivalent matrix, (ii) Singular matrix, (iii) Trace, (iv) Symmetric matrix.
(b) Point out three cases with example that matrices do not follows arithmetic operations as real numbers. (09)
(c) Test linear dependence and independence of the following vectors. If possible express the vectors as a linear combinations. Hence verify the linear combination (if exist); where, [0, 1, 2, 3], [2, 8, 9, 1], [2, 6, 5, -5]. (14)

2. (a) Reduce the following matrix to echelon form, canonical form and normal form. Hence find its rank, $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 2 & 1 & 2 & 1 \\ 4 & 2 & 0 & 2 \end{bmatrix}$. (17)
(b) Is the following matrices are invertible, why? $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $\begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$. (06)
(c) If possible find the inverse of A. Hence express A as well as A^{-1} (if possible) as a product of elementary matrices; where, $A = \begin{bmatrix} 0 & 1 \\ 2 & 4 \end{bmatrix}$. (12)

3. (a) What is the value of λ so that the following system of linear equations has (i) Unique solution, (ii) Many solution, (iii) No solution? where, $\begin{matrix} \lambda x + y = 2 \\ x + \lambda y = 2\lambda \end{matrix}$ (17)
(b) Find the largest Eigen value and corresponding Eigen vector of the Markov matrix, $A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$. Hence verify. Also find the largest Eigen value of A^{10} and corresponding Eigen vectors (by using property). (18)

4. (a) Write down three properties of Eigen value. (06)
(b) Using Laby-Hamelton theorem find the value of A^2 and A^{-2} (if possible) of the matrix, $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$. (09)
(c) Define trivial and non-trivial solution of a set of homogeneous equations. Find the non-trivial solutions of $4x_1 - x_2 + 2x_3 + x_4 = 0$, $2x_1 + 3x_2 - x_3 - 2x_4 = 0$, $7x_2 - 4x_3 - 5x_4 = 0$, $2x_1 - 11x_2 + 7x_3 + 8x_4 = 0$. (20)

Section – B

5. (a) Find the velocity and acceleration of a particle which moves along the curve $x = 2\sin 3t$, $y = 2\cos 3t$, $z = 8t$ at any time $t > 0$. Also find the magnitude of the velocity and acceleration along x axis at $t = \frac{\Pi}{6}$. (13)

(b) If $\frac{d^2\bar{q}}{dt^2} = 12t\hat{i} + 24t^2\hat{j} + \sin t\hat{k}$ such that $\bar{q}(0) = \hat{i} + \hat{j}$ and $\frac{d\bar{q}}{dt} = 2\hat{i} + 3\hat{j}$, $t = 0$. Then, find the value of $\bar{q}$ at any t, and at $t = 5$. (16)

(c) What are the physical meaning of ∇f and $\nabla \cdot \bar{F}$? (06)

6. (a) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2,-1,2)$. (11)

(b) Show that $\bar{F} = (y^2 + 2xz^2)\hat{i} + (2xy - z)\hat{j} + (2x^2z - y + 2z)\hat{k}$ is irrotational and hence find its scalar potential. (13)

(c) Calculate $\int_C \bar{F}.d\bar{r}$, where C is the port of the spiral $\bar{r} = <a\cos\theta,\ a\sin\theta,\ a\theta>$ corresponding to $0 \le \theta \le \frac{\Pi}{2}$ and $\bar{F} = r^2\hat{i}$. (11)

7. (a) Evaluate $\iint_S \bar{F}.\hat{\eta}ds$, where $\bar{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ over the surface of $x^2 + y^2 + z^2 = 1$ which lies in the first octant. (12)

(b) For Laplace transform, write following properties: First shifting, Second shifting, Scaling, Multiplication by powers of t, Division by t. Also state the convolution theorem for inverse Laplace transform. (10)

(c) Find the Laplace transform of $\frac{\sin at}{t}$. Does the transform of $\frac{\cos at}{t}$ exist? Justify. (13)

8. (a) Find Laplace transforms of: (i) $t^2 \sin 2t$, (ii) $e^{-4t}\frac{\sin 3t}{t}$, (iii) Unit step function. (11)

(b) Find the inverse Laplace transforms of: (i) $\frac{s^2+3}{s(s^2+9)}$, (ii) $\frac{s}{s^2+4s+13}$, (iii) $\log\frac{s^2-1}{s^2}$. (15)

(c) Using the convolution theorem, find $L^{-1}\left\{\frac{s^2}{(s^2+1)(s^2+2)}\right\}$.
